

```
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-- SQL CLAUSES COMPLETE LAB

-- Covers: SELECT, FROM, WHERE, GROUP BY, HAVING, ORDER BY

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-- =====

-- 1. CREATE DATABASE

-- =====

DROP DATABASE IF EXISTS sql_clause_lab;

CREATE DATABASE sql_clause_lab;

USE sql_clause_lab;


-- =====

-- 2. CREATE TABLE

-- =====

CREATE TABLE sales (

    sale_id INT AUTO_INCREMENT PRIMARY KEY,

    customer_name VARCHAR(100),

    product VARCHAR(100),

    region VARCHAR(50),

    amount DECIMAL(10,2),

    sale_date DATE

);
```

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-- 3. INSERT SAMPLE DATA

-- =====

INSERT INTO sales (customer_name, product, region, amount, sale_date)

VALUES

('Aung Aung', 'Laptop', 'Yangon', 1500, '2025-01-01'),

('Su Su', 'Phone', 'Mandalay', 800, '2025-01-02'),

('Kyaw Kyaw', 'Laptop', 'Yangon', 2000, '2025-01-03'),

('Mya Mya', 'Tablet', 'Bago', 600, '2025-01-04'),

('Ko Ko', 'Phone', 'Yangon', 900, '2025-01-05'),

('Hla Hla', 'Laptop', 'Mandalay', 1800, '2025-01-06');

-- View all data

38 • **SELECT** * **FROM** sales;

Result Grid						
Filter Rows:						
Edit:   						
Export/Import:  						
	sale_id	customer_name	product	region	amount	sale_date
▶	1	Aung Aung	Laptop	Yangon	1500.00	2025-01-01
	2	Su Su	Phone	Mandalay	800.00	2025-01-02
	3	Kyaw Kyaw	Laptop	Yangon	2000.00	2025-01-03
	4	Mya Mya	Tablet	Bago	600.00	2025-01-04
	5	Ko Ko	Phone	Yangon	900.00	2025-01-05
	6	Hla Hla	Laptop	Mandalay	1800.00	2025-01-06
*	NULL	NULL	NULL	NULL	NULL	NULL

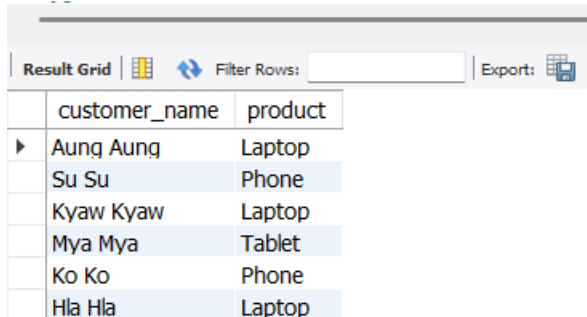
-- =====

-- 4. SELECT + FROM

-- Select specific columns from table

-- =====

```
44 • SELECT customer_name, product
45 FROM sales;
```



	customer_name	product
▶	Aung Aung	Laptop
	Su Su	Phone
	Kyaw Kyaw	Laptop
	Mya Mya	Tablet
	Ko Ko	Phone
	Hla Hla	Laptop

-- =====

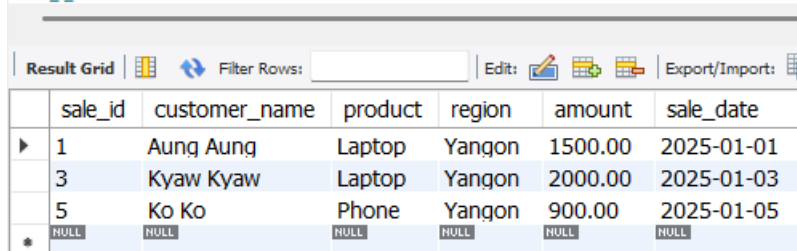
-- 5. WHERE CLAUSE

-- Filter rows based on condition

-- =====

-- Filter by region

```
53 • SELECT * FROM sales
54 WHERE region = 'Yangon';
--
```



	sale_id	customer_name	product	region	amount	sale_date
▶	1	Aung Aung	Laptop	Yangon	1500.00	2025-01-01
	3	Kyaw Kyaw	Laptop	Yangon	2000.00	2025-01-03
	5	Ko Ko	Phone	Yangon	900.00	2025-01-05
*	NULL	NULL	NULL	NULL	NULL	NULL

-- Multiple conditions

```
57 • SELECT * FROM sales
58 WHERE amount > 1000 AND region = 'Yangon';
59
```

Result Grid						
		Filter Rows:			Edit:	Export/Import:
	sale_id	customer_name	product	region	amount	sale_date
▶	1	Aung Aung	Laptop	Yangon	1500.00	2025-01-01
	3	Kyaw Kyaw	Laptop	Yangon	2000.00	2025-01-03
*	NULL	NULL	NULL	NULL	NULL	NULL

-- =====

-- 6. GROUP BY CLAUSE

-- Group rows and apply aggregation

-- =====

-- Total sales by region

```
66 • SELECT region, SUM(amount) AS total_sales
67 FROM sales
68 GROUP BY region;
```

Result Grid		
		Filter Rows:
		Export:
		Wrap Cell Content:
	region	total_sales
▶	Yangon	4400.00
	Mandalay	2600.00
	Bago	600.00

-- Count orders per product

```
71 • SELECT product, COUNT(*) AS total_orders
72 FROM sales
73 GROUP BY product;
74
```

Result Grid		
		Filter Rows:
		Export:
		Wrap Cell Content:
	product	total_orders
▶	Laptop	3
	Phone	2
	Tablet	1

-- =====

-- 7. HAVING CLAUSE

-- Filter grouped results

-- =====

-- Regions with total sales > 2000

```
81 • SELECT region, SUM(amount) AS total_sales
82 FROM sales
83 GROUP BY region
84 HAVING SUM(amount) > 2000;
```

Result Grid		
Filter Rows:		
Export: Export		
Wrap Cell Content: Wrap		
	region	total_sales
▶	Yangon	4400.00
	Mandalay	2600.00

-- =====

-- 8. ORDER BY CLAUSE

-- Sort results

-- =====

-- Sort by amount ascending

```
92 • SELECT * FROM sales
93 ORDER BY amount ASC;
```

Result Grid						
Filter Rows:						
Edit: Edit						
Export/Import: Export						
	sale_id	customer_name	product	region	amount	sale_date
▶	4	Mya Mya	Tablet	Bago	600.00	2025-01-04
	2	Su Su	Phone	Mandalay	800.00	2025-01-02
	5	Ko Ko	Phone	Yangon	900.00	2025-01-05
	1	Aung Aung	Laptop	Yangon	1500.00	2025-01-01
	6	Hla Hla	Laptop	Mandalay	1800.00	2025-01-06
	3	Kyaw Kyaw	Laptop	Yangon	2000.00	2025-01-03
*	NULL	NULL	NULL	NULL	NULL	NULL

-- Sort by amount descending

```
96 • SELECT * FROM sales
97 ORDER BY amount DESC;
```

Result Grid						
		Filter Rows:		Edit:		Export/Import:
	sale_id	customer_name	product	region	amount	sale_date
▶	3	Kyaw Kyaw	Laptop	Yangon	2000.00	2025-01-03
	6	Hla Hla	Laptop	Mandalay	1800.00	2025-01-06
	1	Aung Aung	Laptop	Yangon	1500.00	2025-01-01
	5	Ko Ko	Phone	Yangon	900.00	2025-01-05
	2	Su Su	Phone	Mandalay	800.00	2025-01-02
	4	Mya Mya	Tablet	Bago	600.00	2025-01-04
*	NULL	NULL	NULL	NULL	NULL	NULL

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-- 9. COMBINED QUERY (REAL USE CASE)

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-- Total sales per region with conditions and sorting

```
104 • SELECT
105     region,
106     COUNT(*) AS total_orders,
107     SUM(amount) AS total_sales
108 FROM sales
109 WHERE amount > 500
110 GROUP BY region
111 HAVING SUM(amount) > 1500
112 ORDER BY total_sales DESC;
```

Result Grid			
		Filter Rows:	Export:
	region	total_orders	total_sales
▶	Yangon	3	4400.00
	Mandalay	2	2600.00

-- =====

-- 10. EXECUTION ORDER (IMPORTANT NOTE)

-- =====

-- SQL Execution Order:

-- 1. FROM

-- 2. WHERE

-- 3. GROUP BY

-- 4. HAVING

-- 5. SELECT

-- 6. ORDER BY

-- =====

-- 11. CLEANUP (OPTIONAL)

-- =====

-- DROP DATABASE sql_clause_lab;